

THERMACART · DESIGN DECISION RECORD

ThermaCart recommendations accepted

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DOCUMENT CONTROL

Document ID	TCT-DDR-002	Version	0.2
Status	Draft	Date	2026-10-02
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/thermacart		
Source	docs/decisions/0002-recommendations-accepted.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-02	Amish Chadha	01 to 03 decided by Amish on 2026-10-02 (TCT-DEC-001, items 4 to 6)
0.1	2026-09-25	Amish Chadha	Record the recommendations accepted by Amish on 2026-09-25, what changed in the repo, and the items still open

0002: Recommendations accepted

- **Date:** 2026-09-25
- **Status:** accepted. The front matter status stays Draft because the documentation standard allows only Draft, In review, Released or Superseded.

■ Context

The TRL 3 session recorded items D1 to D8 and T1 to T4 in TCT-DDR-001 as "adopted as recommended for TRL 3 under Amish's 2026-09-25 instruction, open for his review", and items O1 to O9 as "Proposed, awaiting Amish" (docs/REVIEW.md , session 2026-09-25, TRL 3). On 2026-09-25 Amish wrote in chat: "i accept all your recommendations, go with them across all repos." Every item that carries a recommendation is therefore decided, and where a recommendation named one of several options, that option is the decision. Items without a recommendation stay open.

■ Decision

Table 1. Items decided by Amish, 2026-09-25: go with recommendation.

#	Item	Decision	What changed in the repo
D1	Envelope	Stock 6 x 2 x 1/8 in tube in a GN 1/3 envelope; WHO PQS 0.6 L size later; extrusion only at volume	Status wording only; the design already followed it
D2	Grades	C5, C25 and H70 paraffins, W0 produce-only option, C5 first	Status wording only
D3	Pitch	"Cold chain, comfort and hot holding"	Already applied to <code>project.yaml</code> and <code>README.md</code> ; no further change
D4	Keying and indicator	Grade nose key, published key table, sight window	Status wording only
D5	2 mm wall	Evaluated and not adopted	Status wording only
D6	Budget	Keep <code>budget_usd</code> at \$80 for one C5 cartridge and one frame	<code>budget_usd</code> unchanged at \$80
D7	First host type	A produce cooler user through a local partner	Status wording only; no partner named (O1)
D8	Fill and shell	Fill as liquid at the grade limit, seal for life, aluminium shell, no HDPE	Status wording only
T1 to T4	TRL 3 refinements	Folding bail on a thermal break, cap stack from plate, stock fin bar, 85 °C oven limit	Status wording only
O4	R5, PCM fraction	Option (a): relax R5 to 38 % for the stock-tube build, with (c) 50 % kept as the target for a later extrusion	R5 in TCT-REQ-001 v0.4 restated; status from not met (38.3 to 39.9 % against 50 %) to met, thin (38.1 to 39.7 % against 38 %)
O5	R10, cold hold	Option (a): dark finish plus user guidance to lay produce on the cartridges; R10 kept at 8 h until a test shows the contact effect	Guidance added to TCT-PRC-001 v0.4 (How it works); R10 target unchanged; hold 1.9 h (bare, v0.1 basis) to 2.4 h (dark, now the design case), still not met. The measurement is TRL 4 work: decided but on hold
O6	R16 and budget	Option (b): get supplier quotes first; do not raise the budget until quotes are in	<code>budget_usd</code> stays \$80. Getting quotes is a purchasing step: decided but on hold with TRL 4. R16 stays not met
O7	R14, H70 from an oven	Adopt a handling rule on the label (oven gloves or a 12 min wait) and prefer pad charging	R14 restated with the rule; status from at risk to met with the handling rule; H70 label text, precis, safety text and drawing note updated
O8	Surface finish	Dark finish	Row 1 of <code>bom/bom.csv</code> adds etch primer and matte black high-temperature paint (\$24 to \$29); <code>finish</code> parameter in <code>cad/src/model.py</code> ; TCT-DWG-001 Rev P1 to P2 with a finish note; media re-rendered with a black shell; TCT-CAL-001 v0.2 uses emissivity 0.9 as the design case and adds 15 g of paint
O9	R1, C25 band		R1 restated; status from at risk to met

#	Item	Decision	What changed in the repo
		Widen the C25 band to 22 to 27 °C	

■ Effect on the numbers

Table 2. Numbers before and after (TCT-CAL-001 v0.1 to v0.2).

Quantity	Before	After
Cartridge parts cost	\$71.00	\$76.00
One C5 cartridge and one frame, against the \$80 budget	\$87.00 (\$7 over)	\$92.00 (\$12 over)
Three-grade set plus one frame	about \$230	about \$245
Empty cartridge	1.78 kg	1.80 kg
Filled mass	2.89 to 2.96 kg	2.90 to 2.98 kg (R4 still met, thin)
PCM fraction	38.3 to 39.9 %	38.1 to 39.7 %
R10 hold, two C5 at 32 °C (design case)	1.9 h bare	2.4 h dark
R11 hold, one H70 (design case)	7.2 h bare	13.6 h dark
R12 C5 freezer, C25 refrigerator, H70 oven (design case)	5.6, 6.6 and 6.1 h	4.2, 4.6 and 5.3 h
Requirement counts	3 not met, 3 at risk, 3 not verifiable, 9 met	2 not met (R10, R16), 1 at risk (R12), 3 not verifiable (R6, R7, R17), 12 met

■ Items still open

Table 3. Items with no recommendation at this record; decided by Amish on 2026-10-02 (TCT-DEC-001, items 4 to 6).

#	Item
01	The named co-design partner and region for the first produce cooler host (Decided by Amish, 2026-10-02 (TCT-DEC-001, item 4): a produce cooler user through a postharvest extension partner; first candidate to approach, the UC Davis Postharvest Technology Center. Nothing is agreed.)
02	Whether ColdPod or ZeerBox should declare ThermaCart as a dependency (Decided by Amish, 2026-10-02 (TCT-DEC-001, item 5): neither declares ThermaCart as a dependency now.)
03	C5 charging: freezer with a conditioning step, or a refrigerator (TCT-CAL-001 shows 31 to 51 h in a refrigerator, which favours the freezer, but no recommendation was made) (Decided by Amish, 2026-10-02 (TCT-DEC-001, item 6): domestic freezer followed by the conditioning step, with the conditioning time printed on the C5 label as a required step.)

■ Consequences

- Controlled documents bumped: TCT-PRB-001 v0.4, TCT-PRC-001 v0.4, TCT-REQ-001 v0.4, TCT-CAL-001 v0.2, TCT-DDR-001 v0.2. Drawing TCT-DWG-001 is at Rev P2.
- `project.yaml`: `budget_usd` stays at \$80, `trl`: 3 and `trl_target`: 3.
- No cross-repo action follows from these decisions; O2 has no recommendation and no other repo was changed.
- The R10 contact measurement and supplier quotes are TRL 4 work, on hold by Amish's instruction. Nothing in this record authorizes building, testing or purchasing.